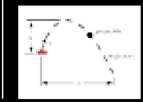

Range of projectiles
Ever wondered what is the maximum range that a projectile can reach? Read this in-depth account to know: **http://dgit.in/Pro_Range**

The math behind food pairing
Head here to know the mathematics that go behind pairing foodstuffs in any typical Indian cuisine: **http://dgit.in/Food_Math**

Mass Driver - The Space Catapult

Just the word "catapult" makes any idea seem silly. But the catapult is an ingenious example of design and engineering. Kids use rubber bands in a catapult to achieve high directional velocity by using the potential energy of the elastic to power their projectiles. So it would seem natural to attempt to do the same for spacecraft. All that would be needed is a seriously large catapult capable of launching thousands of tons of payload at escape velocities. Clearly those would be some really large rubber bands. But humanity is better than that. And thus the mass driver space launch technology was created.

Mass drivers are electromagnetically powered linear motor accelerators intended to be used for non-rocket based space launches. They are similar in principle to rail guns and maglev trains - they use electromagnetic pulses to accelerate their payload. The idea when it was built as a prototype in 1976 by Gerard O'Neill was a proof of concept. However, when researchers began envisioning its scalability for space launches they encountered numerous problems, the least of which was human death. The physics of mass drivers is identical to that of catapults, which means that the projectile launched undergoes tremendous amounts of G-forces - enough to liquify a human being. In addition to which the effectiveness of its power diminishes with larger masses. While for the moment any plans for space catapult are on hiatus, it isn't inconceivable that we might end up using them on non-Earth bodies where gravity and atmospheric restrictions might actually allow it to work.

The G-force generated riding a space catapult would kill any human in seconds.

Project Icarus

It is always creepy when science fiction and science fact converge towards the future. Project Icarus was announced in 2009 by Kelvin Long and Richard Obousy at the British Interplanetary Society. The core idea behind Project Icarus was to successful attempt an interplanetary voyage for humans. Ambitious and noble in its goals, the project did have some very interesting attributes, such as its using nuclear fusion powered engines and using planetary bodies like gaseous planets as refueling stations. And while it was conceived as an unmanned interstellar space probe mission it did have a creepy payload - human embryos.

As science fictiony as this idea may seem it would solve the issue of having humans survive a generational voyage without going extinct or crazy in a sealed metal tube.

Raising embryos to adulthood on these journeys would require child rearing robots.

Space Elevator

Anyone who remembers reading Roald Dahl's Charlie and the Chocolate Factory would remember the TARDIS like glass elevator that shoots off into space. The idea of the space elevator is thankfully not as fantastical. It is in fact far stranger, considering how accurately it fits in with physics as we know it. The design calls for three simple elements – a base station, a counterweight and a connecting tether. The counterweight would need to be the size of a small asteroid or space station that is connected to the base station by a high tensile tether. Once the space elevator is established we know that the Earth's rotation would exert great upward centrifugal force on the counterweight keeping the tether taut and firm like a building. We could then attach "climber" chambers that would then be used to travel up and down from orbit.

The idea of a space elevator traces it's origins back as far as 1895 when Konstantin Tsiolkovsky proposed its construction. Initially it was designed as a freestanding structure which would extend from the surface of the Earth to altitudes of a geostationary orbit. Since then it was radically evolved to being a tether design. In the last ten years alone the interest in space elevators has soared with institutions like NASA and Google supporting research. With breakthroughs in high tensile material technology coming every year this interest isn't surprising. Using carbon nanotubes a Japanese corporation estimated that it could construct a space elevator with as little as USD 5 billion.

Anti-Gravity Drive

The name itself is enough to make the anti-gravity drive understood by everyone.